3. If U={a, b, c, d, e, f, g, h}, A={b, d, f, h} and B={a, d, e, h}, find the following sets.

(i) A' (ii) B' (iii) $A' \cup B'$ (iv) $A' \cap B'$ (v) $(A \cup B)'$

(vi) $(A \cap B)'$ (vii) $(A')'$ (viii) $(B')'$

4. Let U={0, 1, 2, 3, 4, 5, 6, 7}, A={1, 3, 5, 7} and B={0, 2, 3, 5, 7}, find the following sets.

(i) A' (ii) B' (iii) $A' \cup B'$ (iv) $A' \cap B'$ (v) $(A \cup B)'$

(vi) $(A \cap B)'$ (vii) $(A')'$ (viii) $(B')'$

5. Find the symmetric difference between the following sets.

(i) P = {2, 3, 5, 7, 11} and Q={1, 3, 5, 11}

(ii) R = {l, m, n, o, p} and S = {j, l, n, q}

(iii) X = {5, 6, 7} and Y={5, 7, 9, 10}

6. Using the set symbols, write down the expressions for the shaded region in the following

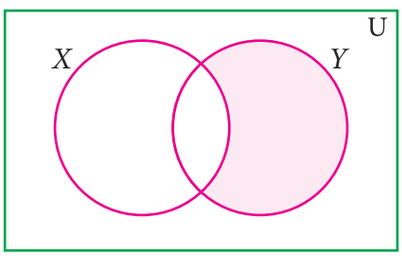


Fig. 1.26

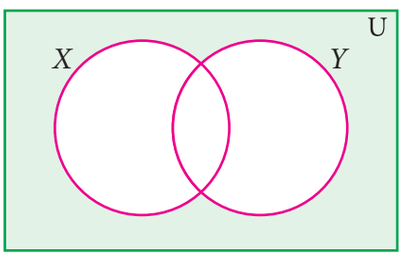


Fig. 1.27

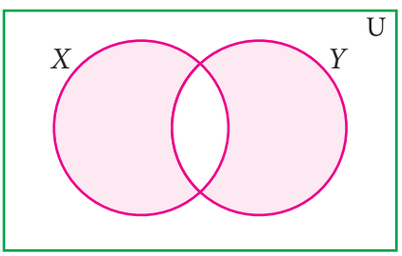


Fig. 1.28

7. Let A and B be two overlapping sets and the universal set be U. Draw appropriate Venn diagram for each of the following,

(i) $A \cup B$ (ii) $A \cap B$ (iii) $(A \cap B)'$ (iv) $(B - A)'$ (v) $A' \cup B'$ (vi) $A' \cap B'$

(vii) What do you observe from the Venn diagram (iii) and (v)?

1.6 Properties of Set Operations

It is an interesting investigation to find out if operations among sets (like union, intersection, etc) follow mathematical properties such as Commutativity, Associativity, etc., We have seen numbers having many of these properties; whether sets also possess these, is to be explored.

We first take up the properties of set operations on union and intersection.

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